

JAWAD ALI

AI Engineer

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CAREER SUMMARY

AI/ML Engineer and BS Artificial Intelligence graduate from COMSATS University Islamabad with a year of hands-on experience building and deploying production AI systems: computer vision and deep learning models (CNNs, OpenCV, PyTorch, TensorFlow), predictive ML and forecasting (XGBoost, LightGBM, time-series), Generative AI and RAG applications (LangChain, LangGraph, FAISS, Pinecone), and automated, event-driven pipelines (n8n, webhooks, scheduled jobs). Ships models behind FastAPI services and Streamlit dashboards, with MLOps practices (MLflow, Docker, CI/CD) on AWS, Azure, GCP and Render. Currently an AI/ML Engineer at Evision Techno Services, delivering a live RAG chatbot and building customer-churn and sales-forecasting systems on company data. Seeking to apply this to real-time, AI-driven products such as automated detection and event-generation systems.

EXPERIENCE

AI / ML Engineer

Sept 2026 – Present

Evision Techno Services

Responsibilities:

- Built and deployed a company FAQ chatbot using Retrieval-Augmented Generation (HuggingFace MiniLM embeddings, FAISS, Llama 3.3 70B via Groq), exposed through a FastAPI endpoint hosted on Render and integrated live with the company server.
- Developing a customer churn prediction system on the company's LPG sales data: leak-free feature engineering, time-based validation, risk scoring with plain-language reasons, a Streamlit dashboard and a FastAPI service (not yet deployed).
- Developing a sales forecasting system (7- and 30-day demand per sale point and cylinder size) with LightGBM, backtested against seasonal baselines, plus customer refill prediction and stock-risk alerts (not yet deployed).
- Designed the data layer for security: read-only access to the ERP database, hashed customer IDs, a separate results database, and role-based API keys.
- Wrote setup guides, system documentation and runbooks so non-technical teams can operate and review the systems.

AI / ML Engineer

Aug 2025 – Aug 2026

OptimAlze BE (Remote, Belgium)

Responsibilities:

- Designed and deployed autonomous LangChain ReAct agents and LangGraph-based multi-agent systems, automating complex multi-step business workflows across cloud infrastructure.
- Built end-to-end AI automation workflows using n8n for intelligent scheduling, data ingestion, webhook handling, routing, and third-party API orchestration.
- Implemented and deployed RAG pipelines integrating FAISS and Pinecone vector databases for scalable semantic search and document-based question answering.
- Developed production-grade deep learning and classical ML pipelines for classification, regression, and NLP tasks; integrated trained models into live systems.
- Connected and fine-tuned LLMs (OpenAI GPT-4, Claude API, HuggingFace Transformers) to CRMs, databases, and operational services.
- Built a handwritten sentence-checking tool using Tesseract OCR for text extraction combined with OpenAI for reading/comprehension enhancement, in Python.
- Built Streamlit demo and evaluation interfaces to validate LLM outputs, RAG accuracy, and agent workflows.
- Enforced MLOps best practices including MLflow experiment tracking, Docker containerization, and CI/CD on AWS and Azure.
- Engineered prompt and function-calling designs to improve LLM reasoning and reduce hallucinations.
- Managed company LinkedIn and supported HR hiring (job posts, CV screening, shortlisting).
- Worked solo end-to-end on all technical deliverables, from architecture design to deployment.

Tools & Technologies:

- **Computer Vision & Deep Learning:** OpenCV, CNNs, RNNs, LSTMs, Transformers, Transfer Learning, TensorFlow / Keras, PyTorch, Image Classification, Object Detection
- **Machine Learning & Forecasting:** scikit-learn, XGBoost, LightGBM, CatBoost, Time-Series Forecasting, Backtesting, Feature Engineering, Churn / Predictive Modelling

- **Generative AI & Agents:** LangChain, LangGraph, ReAct Agents, Tool / Function Calling, Prompt Engineering, Copilot Studio
- **LLMs & Retrieval:** OpenAI GPT-4 / GPT-4o, Claude API, Groq (Llama 3.3), HuggingFace, Sentence Transformers, RAG, FAISS, Pinecone
- **Automation & Real-Time Pipelines:** n8n, Webhooks, Scheduled Jobs, Event-Driven Pipelines, Third-Party API Integration
- **Backend & Data:** Python, FastAPI, Flask, REST APIs, JWT Auth, PostgreSQL, MySQL, MongoDB, Redis, Pandas, NumPy
- **MLOps & Deployment:** Docker, MLflow, CI/CD, AWS (EC2, S3, Lambda), Azure, GCP, Render, Kubernetes, Git / GitHub
- **Frontend & Demos:** Streamlit, React.js, Node.js, Express.js, MERN Stack, Power BI

EDUCATION

COMSATS University Islamabad

2022 – 2026

BS in Artificial Intelligence

Relevant Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Data Science, Statistical Modelling, Neural Networks, Databases, Software Engineering

PROJECTS

FAQ RAG Chatbot (Evision Techno Services)

2026

Stack: Python • LangChain • HuggingFace MiniLM • FAISS • Groq (Llama 3.3 70B) • Streamlit • FastAPI • Render

- Built a chatbot that answers customer questions from the company's FAQ PDFs and CSV, using local embeddings and a FAISS index.
- Kept the RAG engine separate from the UI, then wrapped it in a FastAPI endpoint deployed on Render; it is live and integrated with the company server.
- Built an ingestion script that rebuilds the vector index whenever the FAQ documents change.

Customer Churn Prediction – LPG Sales (Company Dataset)

2026 · In development

Stack: Python • scikit-learn • Pandas • MySQL data • Streamlit • FastAPI • MongoDB • Docker

- Built an end-to-end pipeline that cleans raw sales, delivery, complaint and return tables into one row per customer (recency, order cycle, overdue ratio, spend, complaints, delays).
- Defined churn by looking back at past snapshot dates and training on what happened next; evaluated on the most recent unseen period and compared against a simple "overdue" rule.
- Scores every active customer as High / Medium / Low risk with a plain-language reason; delivered through a Streamlit dashboard (risk lists, Customer 360, model metrics) and a FastAPI service, with nightly scoring and monthly retraining.
- Wrote a full guide covering setup, limitations and a prioritised improvement roadmap. Not yet deployed.

Sales Forecasting & Refill Prediction – Noor Gas

2026 · In development

Stack: Python • LightGBM • Pandas • FastAPI • Streamlit • MySQL • Docker • LLM agents

- Forecasts daily cylinder demand per sale point and cylinder size for 7 and 30 days using a global LightGBM model with calendar features (weekday, winter, Ramadan, Eid) and quantile ranges.
- Added a backtesting framework with an acceptance rule: the model is used only if it beats seasonal-naive baselines on MAE and WAPE; unit tests prove no future data leaks into features.
- Built a per-customer refill model (due, overdue, lapsed), stock-risk alerts from days of cover, and small agents for natural-language questions, daily briefings and draft reminders requiring staff approval.
- Security-first design: read-only ERP access, hashed customer IDs, role-based API keys, rate limiting, audit logs and an approved-function layer so the LLM never writes SQL. Documented in full; not yet deployed.

Car Price Analyst Agent

2025

Stack: Python • LangChain • OpenAI API • XGBoost • Selenium • BeautifulSoup • MongoDB • Streamlit • Cron Jobs

- Built an end-to-end system scraping second-hand car listings from AutoScout24, GoCars, and 2ehands, scaling from an initial 20–30k listings to 200k+ listings.
- Trained and compared XGBoost plus two additional models for price prediction, reaching ~92% accuracy against a 97% target; shipped a rule-based pricing approach for the MVP while the trained ML model remains deployed live in parallel.
- Set up a cron job on a remote server to scrape on a schedule, clean incoming data automatically, and store it in MongoDB, requiring no manual trigger.
- Delivered the user-facing product as a Streamlit app for on-demand vehicle price checks.

Business Analyser Agent

2026

Stack: Python • LangChain ReAct • OpenAI API • Streamlit • MLflow • AWS

- Built an autonomous agent that acts as a personal business analyst for a company, answering conversational questions about its projects and generating business proposal documents.
- Implemented a multi-step LangChain ReAct agent with tool-calling, memory management, and intelligent decision routing.
- Delivered as a Streamlit chatbot combining conversational Q&A and document viewing in one interface.

HEZOP Analyzer Agent

2025

Stack: Python • LangChain • OpenAI API • RAG • FastAPI • Streamlit

- Built for a real client: an AI agent answering process-safety engineering questions using a RAG pipeline over safety documents.
- Converted freelancer-exported AutoCAD DXF files into structured JSON, then built the analysis and RAG pipeline on top of that data.
- Enabled HEZOP engineers to get answers grounded in official documentation.

SKILLS

- **Professional:** Computer Vision, Deep Learning, Predictive Modelling & Forecasting, LangChain, LangGraph, Agentic AI, LLMs / Generative AI, FastAPI, Flask, React.js, Node.js
- **Also:** n8n, Make.com, Gemini, Mistral AI, Ollama, ChromaDB, Elasticsearch, OpenSearch, SQLite, SQL Server, Firebase, NLTK, spaCy, Selenium, BeautifulSoup, OCR (Tesseract)

CERTIFICATIONS

Anthropic AI Courses (May 2026) — Claude 101, Claude Code In Action, Claude With Anthropic API, Intro To Agent Skills, Intro to MCP